Discussion 13: Complex numbers and waves – Solutions

Names:

Section:

Complex numbers are a very useful tool for calculations, but they also unify several of the topics we've discussed so far in class. In discussion this week, we'll see how complex numbers illustrate the close relation between ordinary trig functions and hyperbolic trig functions, provide an alternative parameterization of the group of 2-dimensional rotations, and allow us to describe solutions to the wave equation in a convenient form.

Part 1: Trig functions, ordinary and hyperbolic

Recall the definitions of the hyperbolic trig functions from Discussion 7:

$$\sinh\phi = \frac{e^{\phi} - e^{-\phi}}{2}, \quad \cosh\phi = \frac{e^{\phi} + e^{-\phi}}{2}. \tag{1}$$

1.1. Write the Cartesian forms for $e^{i\theta}$ and $e^{-i\theta}$, and use them to solve for $\sin\theta$ and $\cos\theta$ in terms of $e^{i\theta}$ and $e^{-i\theta}$.

$e^{\pm i\theta} = \cos\theta \pm i\sin\theta$. Adding these two expressions isolates the real part:

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}.$$

Subtracting them isolates the imaginary part, and dividing by $2i$ gives us

$$\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

1.2. Your answer to problem 1.1 should look suspiciously like the hyperbolic trig formulas, but with some extra factors of i. Put the i's in the appropriate place (using $i^2 = -1$ where necessary) to show the following:

$$\cos(iy) = \cosh y, \quad \sin(iy) = i\sinh y. \tag{2}$$

When y is a real number, this lets us *define* the trig functions of an imaginary number, a process known as *analytic continuation.*

Plugging in $\theta = iy$, we have

$$\cos(iy) = \frac{e^{i(iy)} + e^{-i(iy)}}{2} = \frac{e^{-\theta} + e^{\theta}}{2} = \cosh y,$$

and

$$\sin(iy) = \frac{e^{i(iy)} - e^{-i(iy)}}{2i} = \frac{e^{-\theta} - e^{\theta}}{2i}$$

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To recover the expression for sinh, we multiply the right-hand side by $i^2 = -1$, which puts the i in the numerator and changes the relative sign of the terms in the numerator to match the expression for sinh.

1.3. Derive Eq. (2) in a different way, by plugging the argument $z = iy$ into the Taylor series of $\cos z$ and $\sin z$ about $z = 0$. This is also a form of analytic continuation: in complex analysis, we can often *define* functions by their Taylor series, and simply declare that the value of the function at any point in the complex plane is the value of the Taylor series at that point. Recall that the Taylor series for the hyperbolic trig functions are

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots, \quad \cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots \tag{3}$$

$$\sin(iy) = (iy) - \frac{(iy)^3}{3!} + \frac{(iy)^5}{5!} + \cdots = i\left(y + \frac{y^3}{3!} + \frac{y^5}{5!} + \cdots\right) = i \sinh y;$$

note that all the minus signs in the Taylor series for sin became plus signs because the y^3, y^7, etc. terms have a factor of $i^2 = -1$ (we also used $i^{4n} = (i^2)^{2n} = (-1)^{2n} = 1$). Similarly,

$$\cos(iy) = 1 - \frac{(iy)^2}{2!} + \frac{(iy)^4}{4!} + \cdots = 1 + \frac{y^2}{2!} + \frac{y^4}{4!} + \cdots = \cosh y,$$

where all factors of i appear as $(i)^{4n+2} = -1$ or $i^{4n} = 1$.

1.4. Analytic continuation is used all over physics, but one particularly convenient application is the concept of "imaginary time." This is really not as mysterious as it sounds, as we'll shortly see. Recall the definition of the invariant interval:

$$\Delta s^2 = t^2 - x^2 - y^2 - z^2. \tag{4}$$

Let $T = it$. Show that with this "imaginary time" variable,

$$-\Delta s^2 = T^2 + x^2 + y^2 + z^2. \tag{5}$$

So, up to a minus sign, this analytic continuation turns Lorentz dot products into ordinary (Euclidean) dot products in 4 dimensions, and all hyperbolic trig functions in the Lorentz transformation matrices become ordinary trig functions (i.e. boosts turn into rotations). This is often used as a trick to make integrals over 4-vectors easier to compute.

Since $T = it$, we have $t = T/i = -iT$, so $\Delta s^2 = (-it)^2 - x^2 - y^2 - z^2 = -t^2 - x^2 - y^2 - z^2$. Multiplying both sides by -1 gives the desired relation. (This is one reason the "mostly-plus" convention of the metric, which is *not* the one we use in this class, is more convenient in other areas of physics such as string theory and general relativity; in that convention, the analytic continuation to T doesn't come with an additional minus sign.)

Part 2: The group U(1)

The polar form of complex numbers makes clear that there is a very close correspondence between complex numbers and rotations in 2 dimensions. In this part, you will show that unimodular complex numbers (the ones with $|z| = 1$) form a group, called U(1), that is identical to the rotation group SO(2) you studied in Discussion 6.

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2.1. Verify the formula

$$e^{i\theta_1}e^{i\theta_2} = e^{i(\theta_1+\theta_2)} \tag{6}$$

by writing both sides in Cartesian form and using trig identities. This shows that the complex exponential function obeys the same algebraic rules as the ordinary one does, $e^{a+b} = e^a e^b$.

The left-hand side in Cartesian form is

$$(\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2) = (\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) + i(\sin\theta_1\cos\theta_2 + \sin\theta_2\cos\theta_1).$$

Using the standard trig identities for the sum of angles, the real part is equal to $\cos(\theta_1+\theta_2)$ and the imaginary part is equal to $\sin(\theta_1+\theta_2)$. This is exactly what we get when we write the right-hand side in Cartesian form as $\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)$.

2.2. The polar form also lets us divide complex numbers: if $z = x + iy = re^{i\theta}$, show that

$$\frac{1}{z} = \frac{1}{r}e^{-i\theta} \tag{7}$$

by writing both sides in Cartesian form. Remember to use the complex conjugate to "realify" the denominator.

First let's compute $1/z$ in Cartesian form:

$$\frac{1}{z} = \frac{1}{x+iy} = \frac{x-iy}{(x+iy)(x-iy)} = \frac{1}{x^2+y^2}(x-iy).$$

Since $r = \sqrt{x^2+y^2}$ (and thus $r^2 = x^2+y^2$), and $x - iy = re^{-i\theta}$ by complex conjugation, the right-hand side is equivalent to

$$\frac{1}{x^2+y^2}(x-iy) = \frac{1}{r^2}\left(re^{-i\theta}\right) = \frac{1}{r}e^{-i\theta}.$$

2.3. The relations (6) and (7) show that if $r = 1$ (i.e. $|z| = 1$), multiplication and division of complex numbers of the form $e^{i\theta}$ results in another complex number of the same form. In other words, *unimodular complex numbers form a group*, where the group operation is (complex) multiplication. Find the complex number z which corresponds to the identity element, and for a general unimodular number $e^{i\theta}$, find its inverse.

The identity element is simply the complex number 1, since $1 \times e^{i\theta} = e^{i\theta}$. The identity belongs to the group because it's of the form $e^{i\theta}$ with $\theta = 0$. The inverse of $e^{i\theta}$ is $e^{-i\theta}$, since by the result of problem 2.1, $e^{i\theta}e^{-i\theta} = e^{i(0)} = 1$.

2.4. Finally, write down the 2×2 rotation matrix which corresponds to the group element $e^{i\theta}$ (hint: it should obey the same multiplication law as Eq. (6), and it should reduce to the identity element when $\theta = 0$). This establishes the *isomorphism* between the two groups U(1) and SO(2): every element of U(1) is matched with one and only one element of SO(2).

The matrix corresponding to $e^{i\theta}$ is $R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ which is an active counterclockwise rotation by the same angle θ.

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Part 3: Complex wave notation and 4-vectors

In PHYS 212 you learned that the solution to the 1-dimensional wave equation can be written as $h(x,t) = A\cos[k(x-vt)+\delta]$, where k is the wavenumber, v is the wave speed, and δ is a phase that represents the fact that the wave may be displaced from the origin by some arbitrary amount.

3.1. What is the frequency f in terms of v and k? Define the angular frequency as $\omega = 2\pi f$, and show that we can rewrite the wave as $h(x,t) = A\cos(kx - \omega t + \delta)$.

k is the wavenumber which is related to the wavelength λ by $k = 2\pi/\lambda$. Since $\lambda f = v$, we have $f = v/\lambda = kv/2\pi$. The coefficient of t in the argument of the wave is $kv = 2\pi f = \omega$, so $k(x - vt) = kx - \omega t$.

3.2. Show that we can express $h(x,t)$ as the real part of a complex exponential:

$$h(x,t) = \mathrm{Re}[Ae^{i(kx-\omega t+\delta)}] \tag{8}$$

where A is a *real* constant representing the amplitude of the wave. Show also that we can write

$$h(x,t) = \mathrm{Re}[\tilde{A}e^{i(kx-\omega t)}] \tag{9}$$

where we now use a *complex* amplitude $\tilde{A}$. What is $\tilde{A}$ in terms of A and δ? Note that this allows us to absorb the phase into the amplitude, which is often very convenient.

For a complex number $z = x + iy$, $\mathrm{Re}[z] = x$. Since $Ae^{i(kx-\omega_t+\delta)} = A(\cos(kx-\omega t+\delta) + i\sin(kx-\omega t+\delta))$, the real part of this expression is $A\cos(kx-\omega t+\delta)$ which was our original form of h. If we choose $\tilde{A} = Ae^{i\delta}$, using the additive property of the exponential (6), we can absorb the phase δ into the complex number A.

With our knowledge of relativity, the factor $kx - \omega t$ in the exponential looks suspiciously like a Lorentz dot product $-k \cdot x$, where $x^\mu = (t, x, 0, 0)$ and $k^\mu = (\omega, k, 0, 0)$. We would be justified in combining ω and k into a 4-vector as long as we could justify the fact that (at least for photons) they transform like a 4-vector, which we will now do.

3.3. At a given point in spacetime, a wave is characterized only by its amplitude and phase, $h = \mathrm{Re}[Ae^{i\delta}]$. Argue that δ is the same for all inertial observers. The easiest way to do that is to define δ by the value of the wave at $t = x = 0$, and consider two different frames that have their origins synchronized.

Following the hint, let's consider two frames S and S' with synchronized origins, $t = x = t' = x' = 0$. In both frames, the value of the wave is $h(0,0) = \mathrm{Re}[Ae^{i\delta}] = A\cos\delta$. Since the velocity of S' was arbitrary (in fact we never used it), we can conclude that δ is the same in all frames.

3.4. Your result from problem 3.3 shows that the phase of the complex amplitude $\tilde{A}$ is Lorentz-invariant. But since our choice of origin was arbitrary, we could have considered any other point (x,t), in which case the phase of the wave would be $kx - \omega t + \delta$. Argue that this implies that the *entire* phase must be Lorentz-invariant. But since we know that x^μ transforms like a 4-vector, the fact that the phase can be written as a Lorentz-invariant quantity means that k^μ *must* transform like a 4-vector also! Quantum mechanics provides the final piece of the puzzle: the photon's 4-wavevector and 4-momentum are related by $p^\mu = \hbar k^\mu$.

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Suppose we wanted to evaluate the phase at some other spacetime point (t_0, x_0) in S. We could simply define a new phase $\delta' \equiv kx_0 - \omega t_0 + \delta$, and a new frame S' with origin coinciding with (t_0, x_0). Applying the argument from problem 3.4, δ' is Lorentz-invariant. Since t_0 and x_0 are arbitrary, the entire phase must be Lorentz-invariant.

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